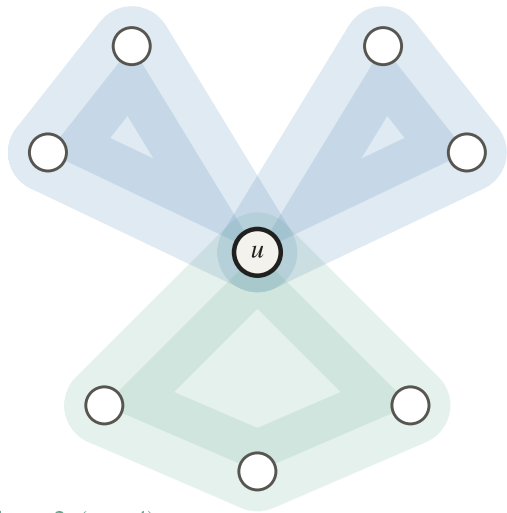


multiplex hypergraph

a

layer 1 ($m_1=3$)



layer 2 ($m_2=4$)

one state per node, shared by every layer

$$k^{(1)}(u) = 2$$

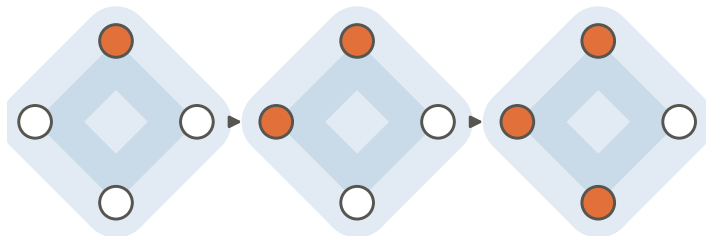
$$k^{(2)}(u) = 1$$

intra-group cascade

b

$$C(m, \lambda, \theta) > (m-1)T$$

each intra-group infection lengthens the clock, so members keep being exposed after the seed has recovered



one seed

it infects a second member

which extends the group's active period

extended

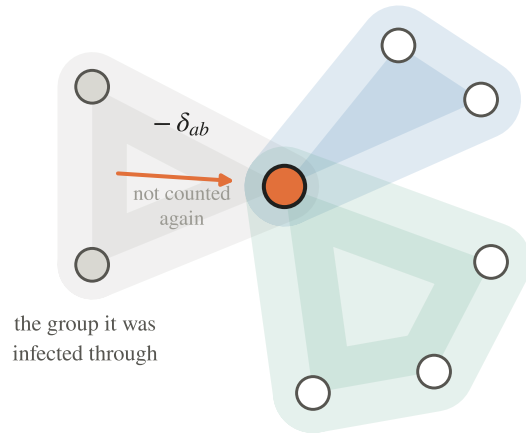


group active while any member is infectious

group-level branching

c

$$C(m_b, \lambda_b, \theta_b) \times \left[\frac{\langle k^{(a)} k^{(b)} \rangle}{\langle k^{(a)} \rangle} - \delta_{ab} \right]$$



the group it was infected through

each remaining group is ignited and yields C new infections: the branching process lives on groups, not nodes